

Submission date: **June 3, 2026**Presentation & Discussion dates: **June 4-5, 2026** (pre-appointed slots per group)

Practical Assignment 2 – Triage Playbook and Framework

Working in groups of up to 4 students, you will analyze this case study and produce a **Triage Playbook and Framework** addressing the requirements presented below.

Scenario Background: The Nexus Logistics Breach

The Target: Nexus Logistics is a global supply-chain provider. Their infrastructure consists of a mixed environment: 300 Windows 11 workstations for corporate staff, and around 20 Ubuntu 22.04 LTS servers handling core business operations (routing APIs, inventory databases, and the customer-facing portal).

The Trigger Incident: at 2:00 AM on a Sunday, the automated Network Intrusion Detection System (NIDS) triggers a high-severity alert. A corporate Windows 11 workstation assigned to a Finance employee (FIN-WKST-42) initiated an anomalous, high-volume SSH connection to the primary Ubuntu production database server (UBUNTU-DB-01).

Shortly after the connection was established, a massive spike in outbound data transfer was detected from UBUNTU-DB-01 to an unclassified external IP address located overseas.

The Constraints:

- **Business Continuity:** The corporate executive team has explicitly stated that shutting down or taking the Ubuntu servers offline will freeze global shipping operations, costing the company 250,000 Euro per hour.
- **Resource Limits:** The incident response team consists of just you and one junior analyst. You do not have the time or storage capacity to take full-disk bit-stream images (*dd* or *.E01*) of every machine on the subnet.
- **The Mission:** You must immediately deploy a **Triage Framework** to rapidly collect volatile evidence, scope the extent of the breach, identify compromised systems, and preserve critical forensic artifacts before they are overwritten or lost.

Your Task

As the Lead Forensics Architect, you must design a comprehensive, dual-OS **Triage Playbook and Framework** tailored to crisis such as this incident. Your document must provide clear, actionable instructions that your junior analyst can execute immediately under high-stress conditions.

Your submission must address both the **Windows 11** and **Ubuntu 22.04** environments based on their unique forensic characteristics and artifacts.

Required Deliverables & Structure

Your final submission must contain the following five sections:

Section 1: Strategic Prioritization & Order of Volatility

- Define your immediate **Order of Engagement**. Which system do you triage first, and why?
- Establish an **Order of Volatility matrix** for both operating systems. Detail specifically which types of data must be collected first to prevent critical evidence destruction.
- Provide a clear business justification for your technical choices, keeping the company's uptime constraints in mind.

Section 2: Windows Workstation Triage Specification (FIN-WKST-42)

Outline the exact forensic artifacts to be targeted on the Windows endpoint without performing a full-disk clone. You must specify:

- **Volatile Data:** How to capture live memory, active network connections, and running processes.
- **Persistence & Execution Artifacts:** Which specific registry hives or Windows artifacts must be gathered to determine how the attacker gained execution.
- **Event Logs:** Identify the precise .evtx log channels required to track lateral movement and authentication anomalies.
- **Tooling:** Name the specific open-source or industry-standard live triage tools you will use to automate this.

Section 3: Ubuntu Server Triage Specification (UBUNTU-DB-01)

Because this server cannot be rebooted or taken offline, your playbook must detail a **Live Linux Triage** methodology:

- **Volatile Data:** Specify the commands required to capture active network sockets, loaded kernel modules (rootkit check), and the running process tree.
- **Log Analysis:** Identify the exact file paths for Linux system logs, authentication logs, and shell histories that will reveal the attacker's commands.
- **User Accounts:** Detail how to check for unauthorized local users or modified privileges.
- **Tooling:** Specify what automation scripts or utilities (e.g., UAC - Unix Artifact Collector, custom bash collection scripts) will be deployed.

Section 4: Supporting Infrastructure

Describe the infrastructure you would propose to support your forensics readiness framework and operations:

- **Support Services and Support Servers:** Identify all required support services and support servers to properly operate your forensics platform. Identify potential key requirements (if any) for those servers.
- **Connection to Servers and Workstations:** Briefly identify how each workstation and each server would use/be supported by the support services and servers identified in the previous point.

Section 5: Data Integrity, Storage, & Chain of Custody

- Detail how the collected triage data will be **transferred off the compromised systems** (e.g., encrypted external drives, or exfiltration to a centralized forensic collection server).
- Explain the exact technical mechanism you will use to **guarantee data integrity** from the moment of collection to final analysis (e.g., cryptographic hashing algorithms).
- Provide a brief template or workflow showing how the **Chain of Custody** will be documented during a rapid live-triage event.

Assessment criteria

Criteria	Description	%
Order of Volatility & Strategy	Correctly prioritizes data collection. Demonstrates a logical, step-by-step progression tailored to the specific breach scenario.	20%
Windows Artifact Selection	Accurately identifies critical Windows forensic locations and matches them with efficient live-collection tools.	15%
Ubuntu Artifact Selection	Correctly identifies Linux-specific forensic locations and matches them with efficient live-collection tools.	15%
Supporting Infrastructure	Correctly identifies the relevant services and servers necessary to deploy and operate the forensics infrastructure, and correctly identifies how they link to servers and workstations.	15%
Business Continuity & Realism	Successfully minimizes operational impact on the production servers. Avoids unnecessary reboots or massive, time-consuming full disk images.	15%
Integrity & Chain of Custody	Outlines a clear method for maintaining evidence custody under pressure.	10%
Clarity & Actionability	The document is professionally formatted, scannable, and written with enough technical clarity that a junior analyst could execute the commands flawlessly.	10%

The **oral presentation of the assignment is mandatory**. It will not be assessed as an isolated parcel, but may directly affect the grading of any of the above-mentioned criteria.

Submission Guidelines

- **Group submission** using **InforEstudiante**.
- **Triage Playbook and Framework** in the form of PDF document, strictly following the standard IEEE double-column conference template (10pt font, specific margins).
- **Optionally**, complementing files organized in a single ZIP.
- **Maximum page limit:** 7 pages.
Highly detailed scripts, raw bash code, or full XML configurations do not belong in the main text. They can be submitted separately within the optional ZIP file and referenced accordingly in the paper.
- **Submission deadline:** **June 3, 2026**
- **Oral presentation & discussion:** **June 4-5, 2026**